Module 8
Static Routing

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Static Routing

Module Objectives:

- Describe static routing and scenarios under which static routing is suitable
- Configure static routes for IPv4 and IPv6 networks
- Perform basic troubleshooting for static route issues

Module References:

- CCNAv7 SRWE – Module 11

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8.1 Introduction to Static Routing

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Introduction to Static Routing

- Recall: Static Routing refers to the technique of manually defining routes to allow routers to forward packets to remote networks

Advantages	Disadvantages
Security is inherent because routers do not transmit network topology information	Does not scale well to large networks because of administrative overhead
No additional route computation (CPU), data structures (memory) or router control messages (network bandwidth) necessary	Does not automatically respond to changes in network topology and traffic conditions
Routes explicitly configured by administrator, therefore network traffic path is predictable	Configuration complexity increases as network size increases

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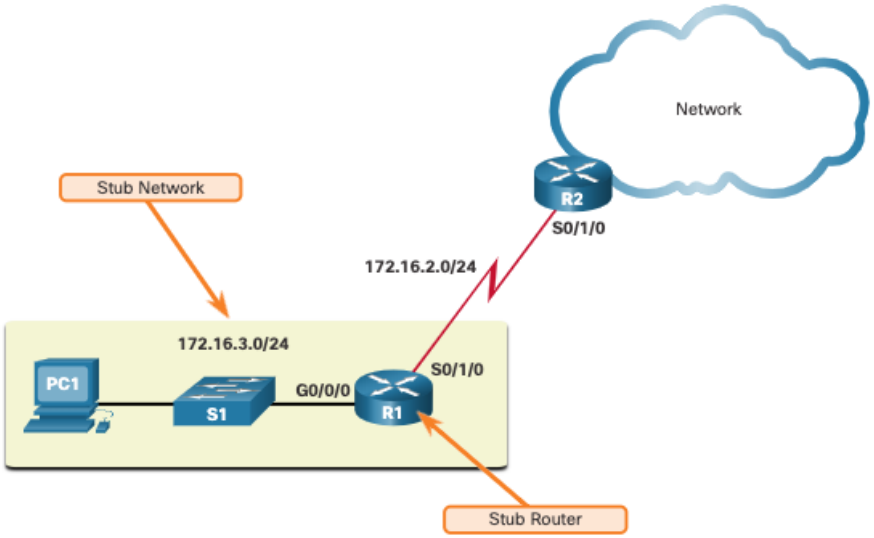
Static Routing Usage Scenarios

- Most networks use a combination of dynamic routing protocols and static routes.
- Static routes are commonly used in the following scenarios:
  - Smaller networks that are not expected to grow
  - As a default route forwarding packets to a service provider
  - For routing between stub networks (network with only one path to an outside network)
  - For routes outside the routing domain and not learned by the dynamic routing protocol
  - When the network administrator wants to explicitly define the path for a specific network



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Types of Static Routes

- Static routes can be configured for IPv4 and IPv6. Both protocols support the following types of static routes:

Standard static route	Default static route
Floating static route	Summary static route

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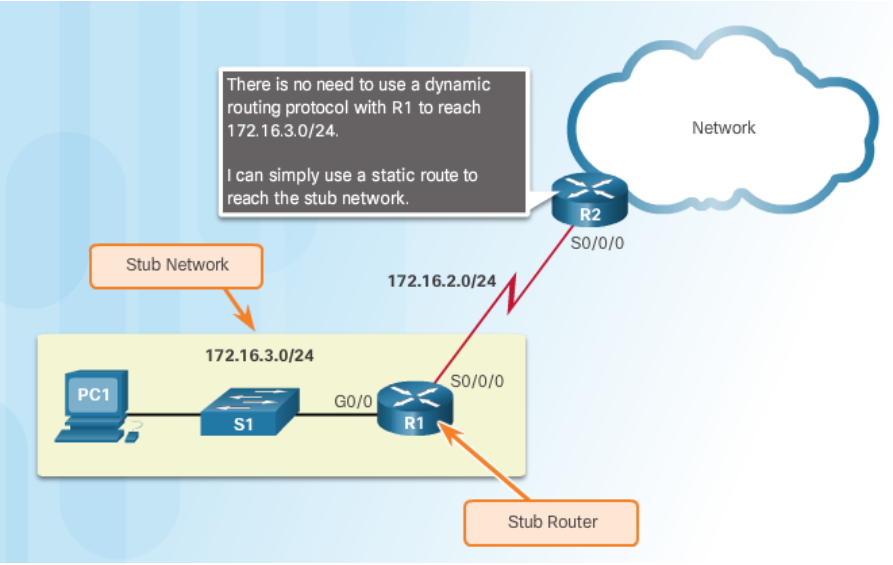
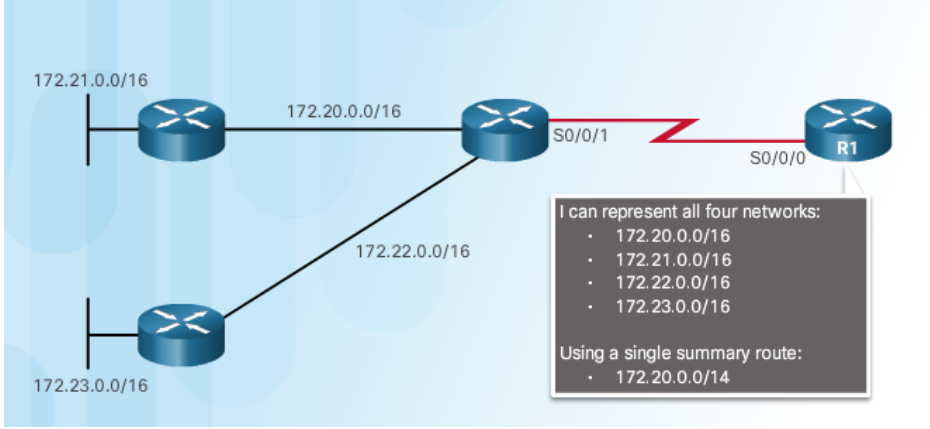
Standard Static Route

- A standard static route is one where the route prefix identifies a specific destination network
- Commonly used on routers serving as neighbors of a stub network



Summary Static Route

- A summary / aggregate route represents multiple routes as a single large network
- Often used in order to reduce the number of routes that need to be maintained in the routing table thereby consuming less RAM and making route lookup faster



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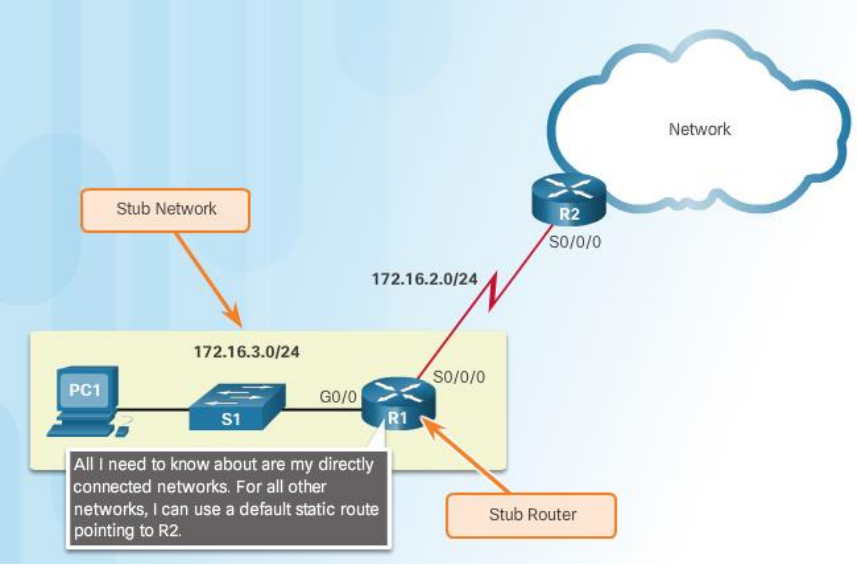
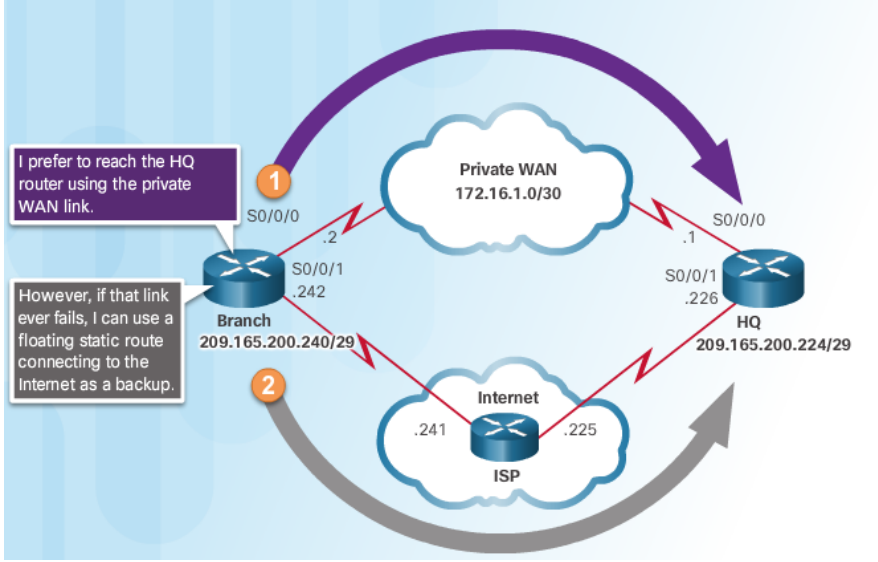
Default Static Route

- Default route is used to match and route all packets that do not match a more specific route in the routing table
- Commonly use on stub routers or edge routers connected to the ISP network



Floating Static Route

- Floating static routes are used to provide a backup path that a router will use when a primary route is not available
- Commonly configured by assigning a higher administrative distance than the main route



8.2 Configuring Static Routes

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IP Route Command

- Static routes are configured using the **ip route** and **ipv6 route** global configuration commands
  - For IPv4: **Router(config)# ip route** *network-address subnet-mask {next_hop | exit_int [next_hop]} [admin_dist]*
  - For IPv6: **Router(config)# ipv6 route** *ipv6-prefix/prefix-length {next_hop | exit_int [next_hop]} [admin_dist]*

Parameter	Description
network-address subnet mask ipv6-prefix/prefix length	Network address of destination e.g. 192.168.1.0 255.255.255.0 2001:db9:acad:1/64
next_hop	IP/IPv6 address of next hop router to reach the destination
exit_int	Egress interface of this router to reach the destination
admin_dist (optional)	Administrative distance assigned to route. AD = 1 if not explicitly specified

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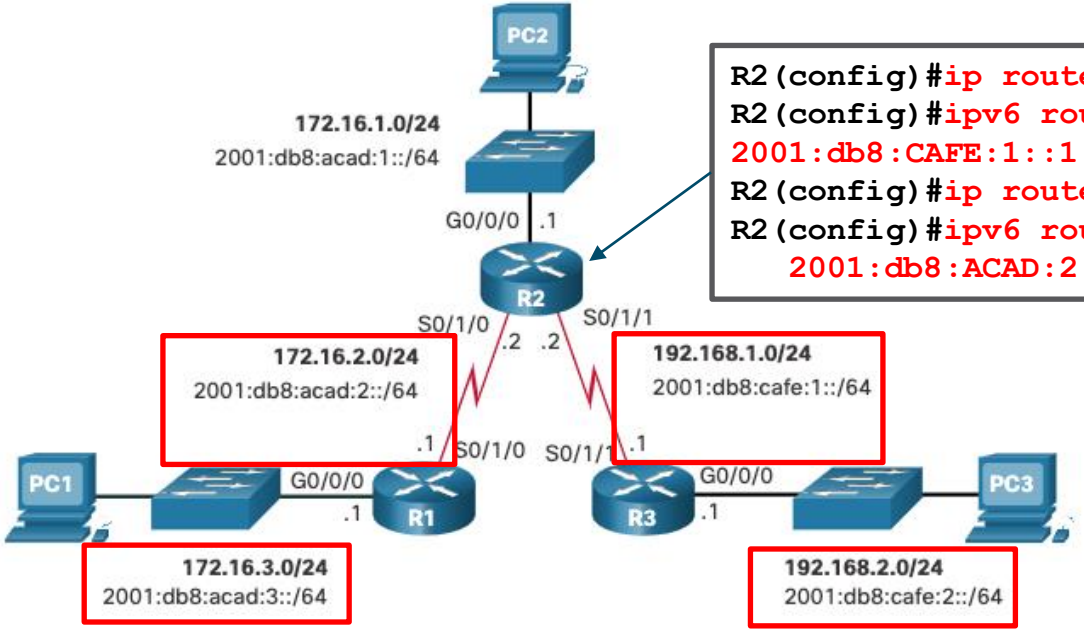
Standard Static Route Configuration

When configuring a static route, the destination can be configured using 3 format options:

- **Recursive route**
  - Only the next-hop IP address is specified.
  - This format is necessary when the exit interface needed is connected to a multi-access network (e.g. Ethernet) where the next hop router needs to be explicitly identified




```
R2(config)#ip route 192.168.2.0 255.255.255.0  192.168.1.1
R2(config)#ipv6 route 2001:db8:CAFE:2::/64
2001:db8:CAFE:1::1
R2(config)#ip route 172.16.3.0 255.255.255.0  172.16.2.1
R2(config)#ipv6 route 2001:db8:ACAD:3::/64
   2001:db8:ACAD:2::1
```

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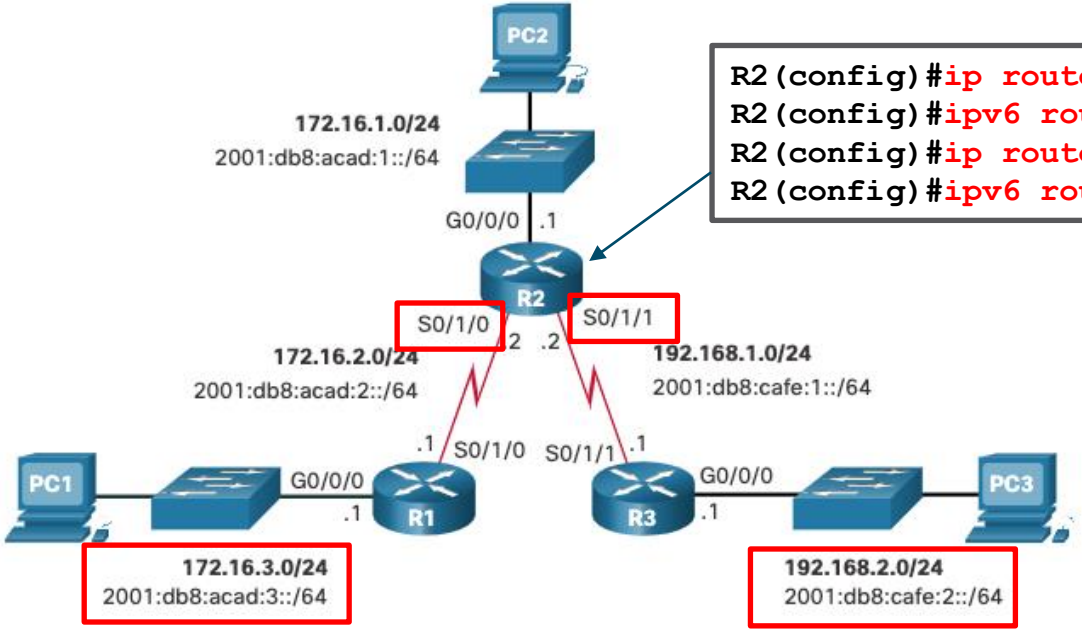
Standard Static Route Configuration

When configuring a static route, the destination can be configured using 3 format options:

- **Directly connected static route**
  - Only the router exit interface is specified
  - Can be used only if the exit interface is connected to a neighbor router using a serial point-to-point connection




```
R2(config)#ip route 192.168.2.0 255.255.255.0  s0/1/1
R2(config)#ipv6 route 2001:db8:CAFE:2::/64  s0/1/1
R2(config)#ip route 172.16.3.0 255.255.255.0  s0/1/0
R2(config)#ipv6 route 2001:db8:ACAD:3::/64  s0/1/0
```

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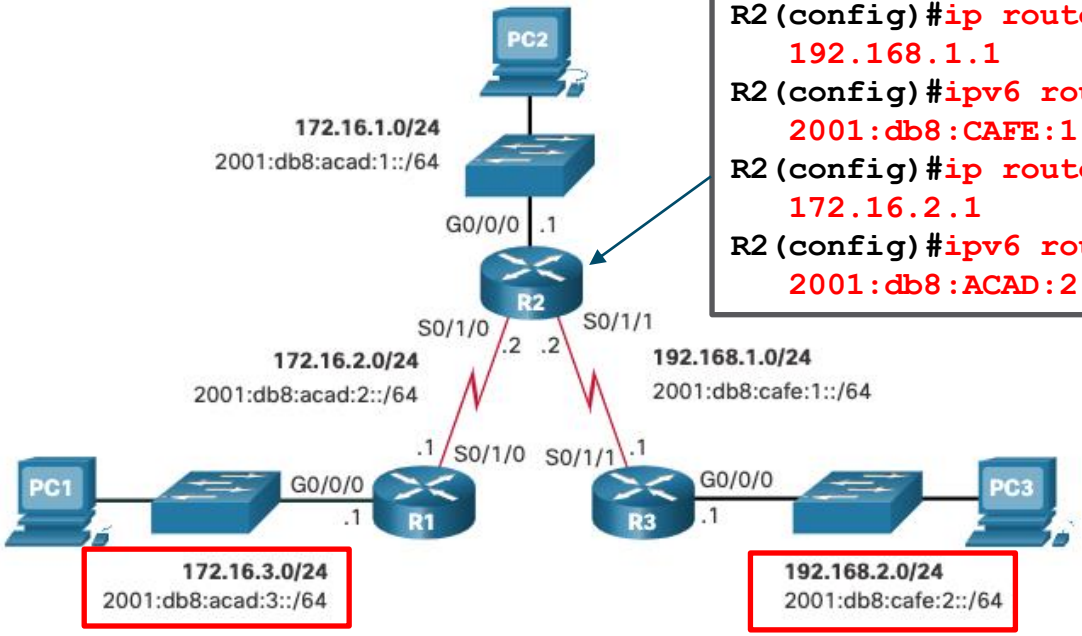
Standard Static Route Configuration

When configuring a static route, the destination can be configured using 3 format options:

- **Fully specified static route**
  - The next-hop IP address and exit interface are specified
  - Mandatory format if using a link-local next hop address for an IPv6 static route




```
R2(config)#ip route 192.168.2.0 255.255.255.0 s0/1/1
   192.168.1.1
R2(config)#ipv6 route 2001:db8:CAFE:2::/64  s0/1/1
   2001:db8:CAFE:1::1
R2(config)#ip route 172.16.3.0 255.255.255.0  s0/1/0
   172.16.2.1
R2(config)#ipv6 route 2001:db8:ACAD:3::/64  s0/1/0
   2001:db8:ACAD:2::1
```

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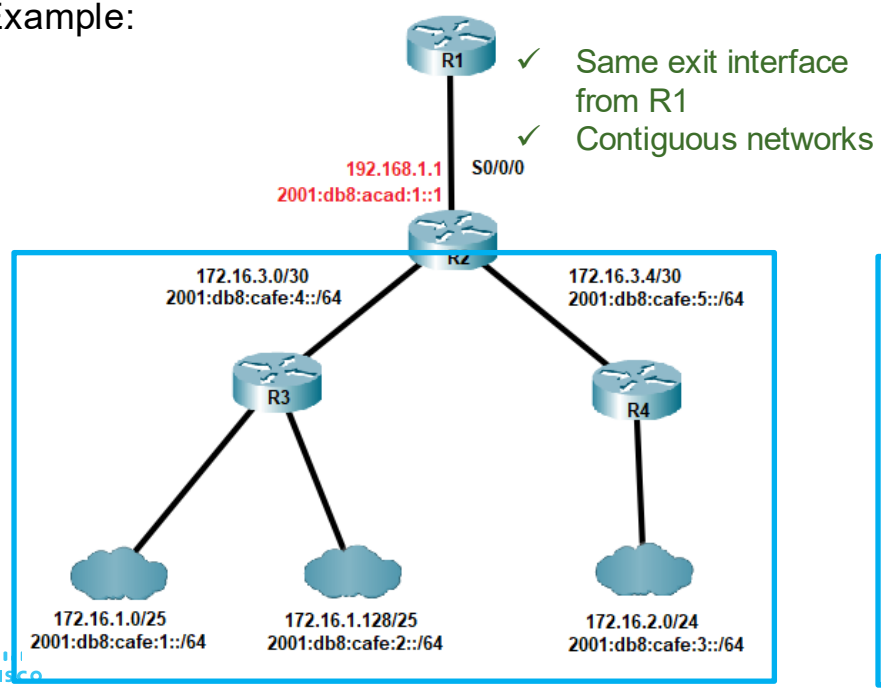
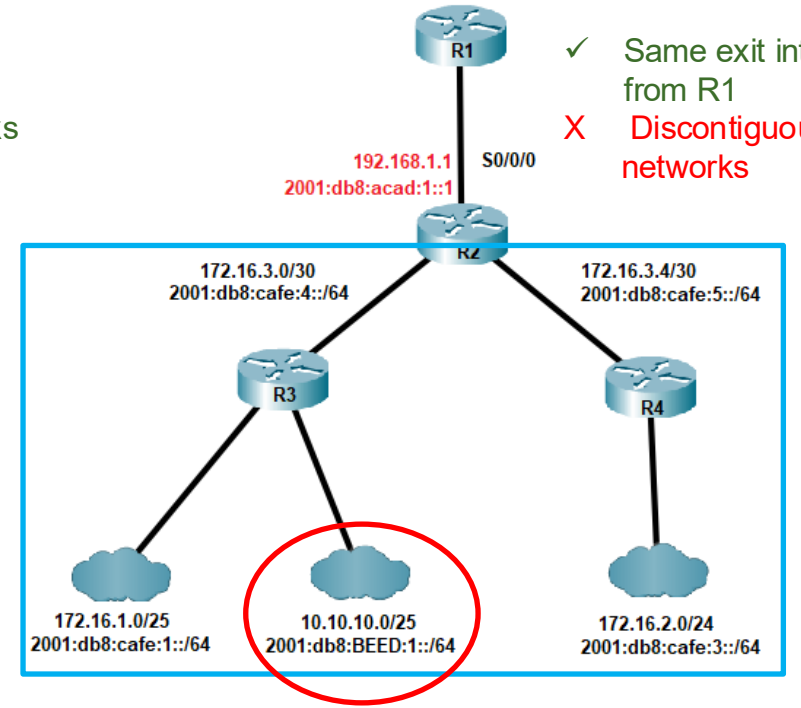
Summary Static Route Configuration

- To create a summary route for multiple subnets, the following conditions must be fulfilled
  - The subnets must be contiguous – i.e. subnets next to each other occupy a sequential address space
  - The route to the subnets must share the same exit interface
  - Example:



✓ Same exit interface from R1

✓ Contiguous networks



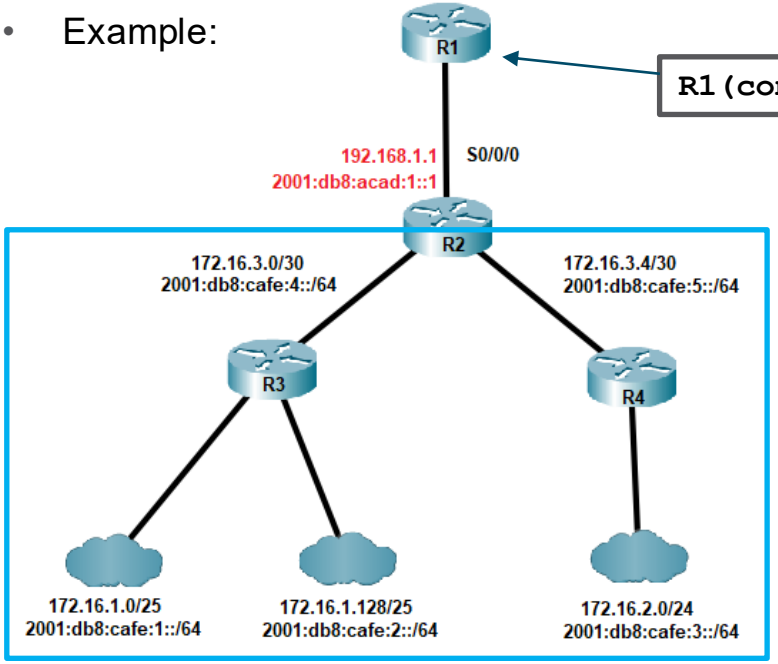
✓ Same exit interface from R1

X Discontiguous networks

Summary Static Route Configuration

- To compute for the summary network address,
  - Step 1: Set the subnet mask / prefix length to where the leftmost bits are common to all subnets
  - Step 2: Retain the common bits
  - Step 3: Zero out bits to the right of the new subnet mask / prefix length
  - Example:

```
R1(config)#ip route 172.16.0.0 255.255.252.0  192.168.1.1
```

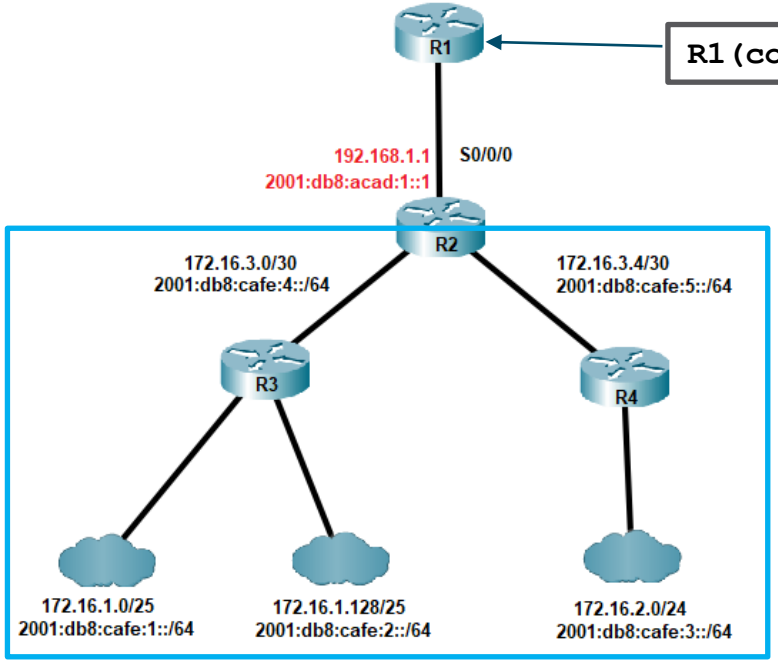


172.16.1.0	10101100.00010000.00000001.00000000
172.16.1.128	10101100.00010000.00000001.10000000
172.16.2.0	10101100.00010000.00000010.00000000
172.16.3.0	10101100.00010000.00000011.00000000
172.16.3.4	10101100.00010000.00000001.00000100
Result	10101100.00010000.00000000.00000000

Summary network = 172.16.0.0/22

Summary Static Route Configuration

- Example (IPv6):




```
R1(config)#ipv6 route 2001:db8:café::/61 2001:db8:acad:1::1
```

2001:db8:café:1::	2001:db8:café:0000 0000 0000 0001: 0….
2001:db8:café:2::	2001:db8:café:0000 0000 0000 0010: 0….
2001:db8:café:3::	2001:db8:café:0000 0000 0000 0011: 0….
2001:db8:café:4::	2001:db8:café:0000 0000 0000 0100: 0….
2001:db8:café:5::	2001:db8:café:0000 0000 0000 0101: 0….
Result	2001:db8:café:0000 0000 0000 0 000: 0….

Summary network = 2001:db8:café::/61

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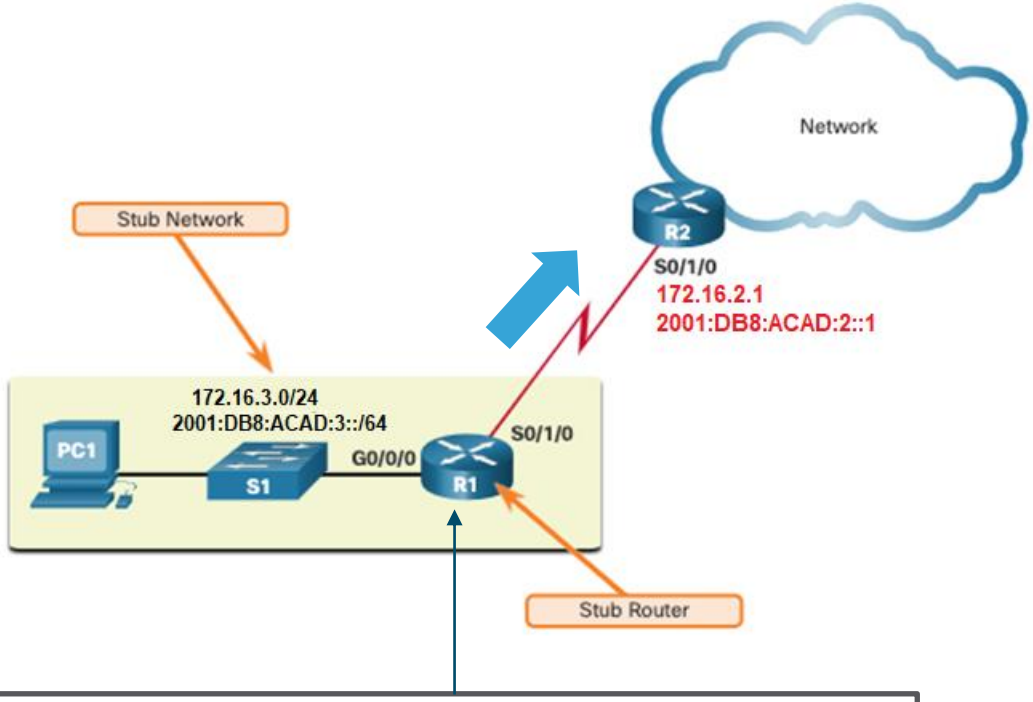
Default Static Route

- A default route is a static route that matches all packets and represents any network that is not in the routing table (a.k.a Gateway of Last Resort)
- Often configured on edge or stub routers
- To define a default route, use an all-0 network address and mask as the route prefix




```
R1(config)#ip route 0.0.0.0 0.0.0.0 172.16.2.1
R1(config)#ipv6 route ::/0  2001:db8:ACAD:2::1
```

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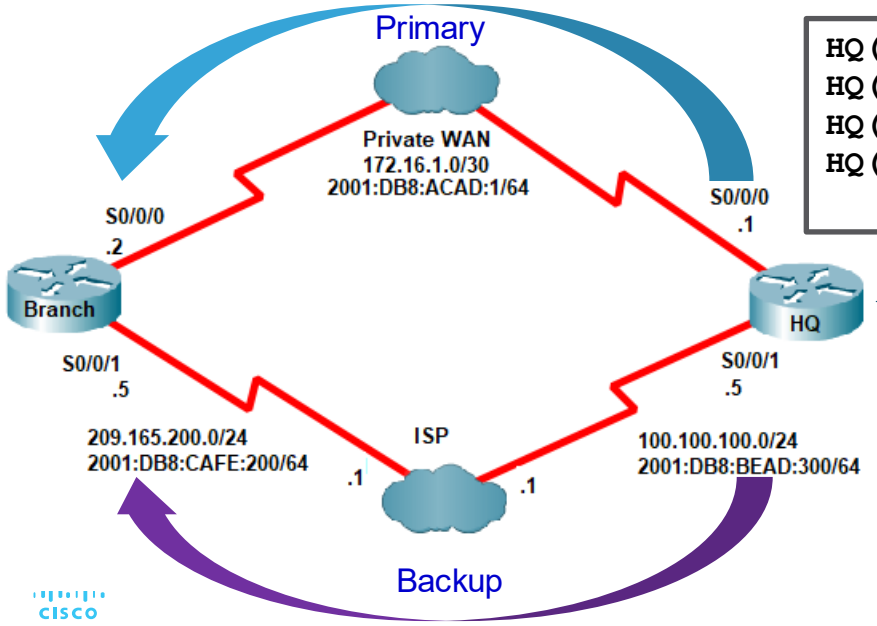
Floating Static Routes

- Floating static are used to provide a backup path to a primary static or dynamic route that will be used only when the primary route is not available
- To accomplish this, the floating static route is configured with a higher administrative distance than the primary route so that becomes less 'trustworthy' than the primary route.




```
HQ(config)#ip route 0.0.0.0 0.0.0.0 172.16.1.2
HQ(config)#ipv6 route ::/0  2001:db8:acad:1::2
HQ(config)#ip route 0.0.0.0 0.0.0.0 100.100.100.1 5
HQ(config)#ipv6 route ::/0  2001:db8:BEAD:300::1 5
```

* Backup uses higher AD

* Default AD of static routes is 1

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Verify Static Routes

show ip route / show ipv6 route – shows all routes available in the routing table

show up route static / show ipv6 route static – shows static routes in the routing table

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.12/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
     192.168.3.0/26 is variably subnetted, 2 subnets, 2 masks
D       192.168.3.0/26 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
D       192.168.3.64/27 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 192.168.1.1, GigabitEthernet0/0
```

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8.3 Troubleshooting Static Route Configuration

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Network Changes

- Networks fail for a number of reasons:
  - An interface can fail
  - A service provider drops a connection
  - Links can become oversaturated
  - An administrator may enter a wrong configuration.
- Network administrators are responsible for pinpointing and solving the problem.
- To efficiently find and solve these issues, it is advantageous to be intimately familiar with tools to help isolate routing problems quickly.

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Common Troubleshooting Commands

Command	Description
`ping`	• Verify Layer 3 connectivity to destination. • Extended pings provide additional options.
`Traceroute / tracert`	• Verify path to destination network. • It uses ICMP echo reply messages to determine the hops to the destination.
`show ip route`	• Displays the routing table. • Used to verify route entries for destination IP addresses.
`show ip interface brief`	• Displays the status of device interfaces. • Used to verify the operational status and IP address of an interface.
`show cdp neighbors`	• Displays a list of directly connected Cisco devices. • Also used to validate Layer 1 and 2 connectivity.

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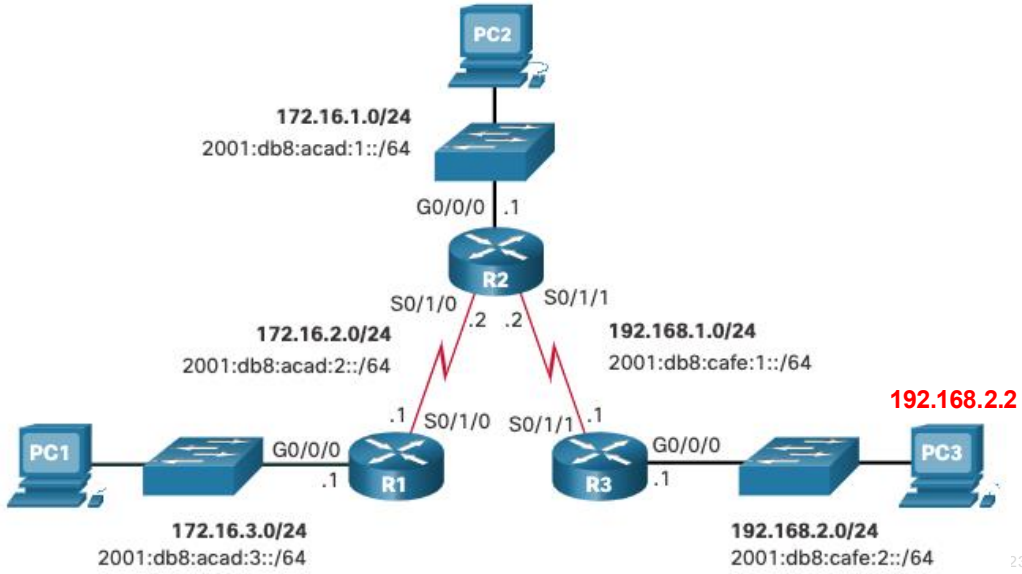
Traceroute Tool

- The traceroute utility is very useful in when diagnosing a routing issue in a network.
  - When using windows, use the tracert command.
  - When performing a trace from a router CLI, use the traceroute command.
- Works by sending successive packets with increasing Time-to-live (TTL) values in order to trace the sequence of hops a packet passes through to reach its destination.

```
C:\WINDOWS\system32>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a maximum
of 30 hops

  1    2 ms     1 ms    1 ms  172.16.3.1
  2    3 ms     2 ms    4 ms  172.16.2.2
  3    5 ms     6 ms    4 ms  192.168.1.1
  4    5 ms     7 ms    7 ms  192.168.2.2

Trace complete.
```



Common Issue 1: Missing or Erroneous Route

- In this scenario, a tracert output shows that the route fails to proceed after the 2nd hop

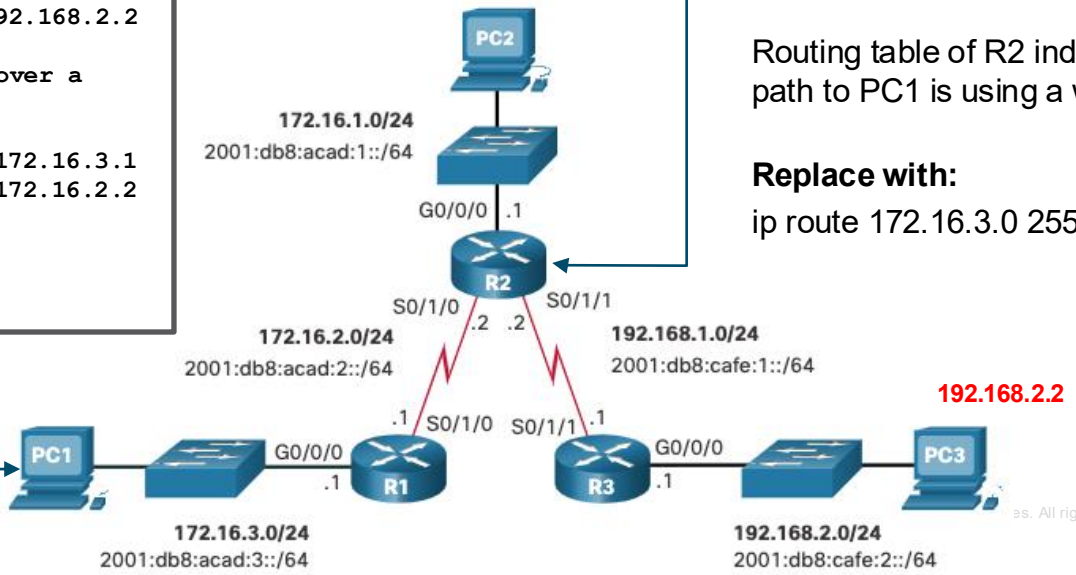
```
R2# show ip route | begin Gateway
Gateway of last resort is not set
      172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C        172.16.1.0/24 is directly connected, GigabitEthernet0/0/0
L        172.16.1.1/32 is directly connected, GigabitEthernet0/0/0
C        172.16.2.0/24 is directly connected, Serial0/1/0
L        172.16.2.2/32 is directly connected, Serial0/1/0
S        172.16.3.0/24 [1/0] via 192.168.1.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, Serial0/1/1
L        192.168.1.2/32 is directly connected, Serial0/1/1
S     192.168.2.0/24 [1/0] via 192.168.1.1
```

```
C:\WINDOWS\system32>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a
maximum of 30 hops

  1    2 ms     1 ms    1 ms  172.16.3.1
  2    3 ms     2 ms    4 ms  172.16.2.2
  3    *        *       *
  4    *        *       *
  5    *        *       *
...
```

Routing table of R2 indicates that the return path to PC1 is using a wrong next hop address.

Replace with:

ip route 172.16.3.0 255.255.255.0 172.16.2.1



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Common Issue 2: Routing Loop

- In this scenario, a tracert output shows that the packet repeatedly bounces between 2 hops until the trace eventually ends

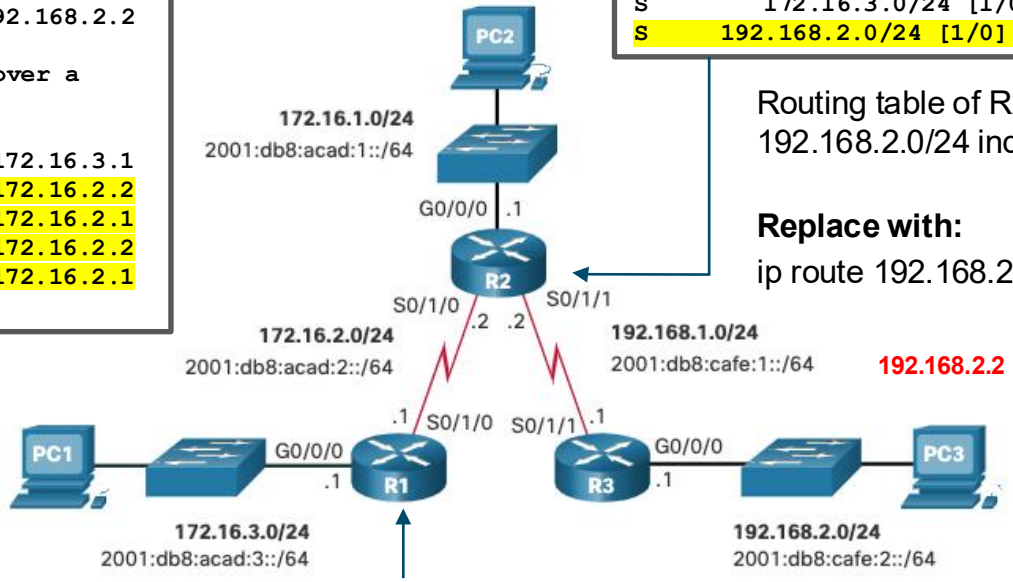
```
C:\WINDOWS\system32>tracert 192.168.2.2

Tracing route to 192.168.2.2 over a
maximum of 30 hops

  1    2 ms    1 ms    1 ms  172.16.3.1
  2    3 ms    2 ms    4 ms  172.16.2.2
  3    4 ms    3 ms    5 ms  172.16.2.1
  4    6 ms    4 ms    5 ms  172.16.2.2
  5    4 ms    3 ms    5 ms  172.16.2.1
...
```

```
R2# show ip route static | begin Gateway
Gateway of last resort is not set
S        172.16.3.0/24 [1/0] via 172.16.2.1
S      192.168.2.0/24 [1/0] via 172.16.2.1
```

Routing table of R2 indicates that the route to 192.168.2.0/24 indicates R1 as its next hop

Replace with:

ip route 192.168.2.0 255.255.255.0 192.168.1.1




```
R1# show ip route static | begin Gateway
Gateway of last resort 172.16.2.2
S*     0.0.0.0/0 [1/0] via 172.16.2.2
```

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Questions?

What did you learn in this module?

- Static routes are manually configured routes that are suitable for small networks or as complement to dynamic routes.
  - Advantages: More secure, less resource utilization, predictable path
  - Disadvantages: Not scalable to large networks, do not automatically respond to topology changes
- Static routes are configured by defining the destination network, the next hop, exit interface and administrative distance
- A standard static route provides a route to a single network
- A summary route combines multiple contiguous subnets sharing the same exit interface from a router into a single route entry
- A default route or gateway of last resort is the route used by packets that do not match any specific network routes in the routing table. It is characterized by an all-0 network address and mask

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What did you learn in this module?

- A floating route serves as a backup route that will be used when a primary route fails. It is configured by setting the administrative distance of the route to be higher than that of the primary route.
- The traceroute command / tool is output the sequence of hops that a packet passes on its way to the destination. It is a useful tool when verifying and troubleshooting route configuration

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